Varun Ghat Ravikumar

AI/ML Engineer
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AI Engineer working in NLP, Linguistics and data pipelines. Specialized in architecting end-to-end LLM pipelines, interactive chatbots, Knowledge Graphs, video processing and document intelligence. Currently working on digitzing etymological dictionaries at ISLE and automated landmarking at LiRI.

Experience

SOFTWARE AND MACHINE LEARNING ENGINEER (ISLE, LiRI) / UNIVERSITY OF ZURICH **FEB 2026 — Present**
Python, LLM, OCR, GCP *Zurich, Switzerland*

- **ISLE**: Working on digitizing etymological databases, orthography normalization, semantic word classification using LLMs and embeddings. *(Oct - Present)*
- **LiRI**: Working on large-scale update of PHOIBLE (cross-linguistic phonetic inventory database) *(Nov - Present)*
- **Previously at LiRI**: Worked on MRI-based mouth/articulator landmark detection models for phonetic articulation research *(Jul - Sep)*
- **Previously at ISLE**: Architected an LLM-driven OCR pipeline to automate data extraction and enhancement for diverse word-list PDF books. Trained models for semantic word classification and diacritic-specific OCR; built tooling for annotation and verification. *(May - Jun)*
- Maintaining GCP infrastructure, supporting data extraction, and contributing to research on abstract concepts and colexification.

RESEARCH ASSISTANT - DDIS GROUP / UNIVERSITY OF ZURICH (Part-time, 20%) **APR 2024 — DEC 2024**
Python, Knowledge Graphs, LLM, GenAI *Zurich, Switzerland*

- Developed an end-to-end pipeline for knowledge graph synthetic dataset generation and augmentation, optimized data ingestion to handle 100 million+ entities efficiently. Implemented synthetic data augmentation in text and images using GenAI.

ASSOCIATE DATA SCIENTIST & INTERN / AMAGI MEDIA LABS **JAN 2023 — AUG 2023**
Python, Video processing, NLP, PyTorch, AWS *Bangalore, India*

- **Fast-tracked from Intern to full-time Data Scientist in just 3 months for high-impact project execution.**
- Engineered a POC sports highlight detection system to automate key moment extraction in soccer videos at 6x real-time speed.
- Deployed a multi-modal genre classification model to enhance categorization in a live production environment.
- Enhanced production pipelines by implementing model drift detection metrics.

INTERNSHIP / KERNEL INSIGHTS **NOV 2022 — JAN 2023**
Stable Diffusion (GenAI), PyTorch *Bangalore, India*

- Fine-tuned Stable Diffusion for consistent storybook stylization. Explored latent seed strategies to control subject poses.

INTERNSHIP / INDIAN INSTITUTE OF ASTROPHYSICS **SEP 2021 — MAY 2022**
C++, C, MPI, PLUTO code *Bangalore, India*

- Migrated C++/MPI modules to PLUTO to implement data-driven MHD simulations of solar flares, studying CME formation.

Education

Master's in Informatics (Major: AI, Minor: DS) **SEP 2023 — MAR 2026**
University of Zurich (*Special student at ETHZ*)
Grade: 5.2 / 6.0

Bachelor of Engineering in Computer Science & Engineering **AUGUST 2017 — MAY 2021**
JSS Science & Technology University (Formerly SJCE), Mysuru
CGPA: 9.01

Publications

- **A Factorial Study of Synthetic Data Generation for Low-Resource Machine Translation using Grammar Books**
  V. G. Ravikumar, S. Ahmadi, L. Jäger, R. Sennrich
  Accepted to the Italian Conference on Computational Linguistics (CLIC-it) 2026 [DOI:10.48550/arXiv.2607.22376]
- **Alan's Speakeasy - An Ecosystem for the Evaluation of Conversational Agents**
  L. Rossetto, C. Sarasua, O. Inel, **V. G. Ravikumar**, et al.
  Proceedings of the ACM Web Conference 2025 (WWW '25), Companion Volume. [DOI: 10.1145/3701716.3715165]

Projects

- **Low-Resource Language Translation using Synthetic Data from Grammar Books** *(LLMs, Python, GCP Vertex AI)* - Master's Thesis
  Designed an automated OCR + NLP pipeline to extract linguistic rules from grammar book PDFs. Generated 190k+ synthetic translation pairs via Gemini. Fine-tuned LLMs on GCP Vertex AI and evaluated 96 experimental setups, outperforming zero-shot baselines.
- **AI Data Analysis Dashboard** *(LLMs, postgreSQL, React)* | *Hack4Good & Caritas*
  Developed an internal analytics tool (chat bot) allowing non-technical employees to visualize and analyze CSV/Excel data via natural language queries. Architected the backend utilizing LLMs and Retrieval-Augmented Generation (RAG) to automate data interpretation.
- **Diacritic-specific OCR (Ongoing)** *(Vision Language Models)*
  Developing an OCR model to improve accuracy on unseen diacritics, with ability to post-correct OCR without finetuning.

Skills

Programming Languages	Python, C, C++, SQL, Javascript, R
Core AI & ML	PyTorch, Scikit-learn, Transformers
NLP	SpaCy, NLTK, Machine translation,LLMs, RAG
Computer Vision	OpenCV, OCR, Diffusion Models, Vision Language Models
Tools	Docker, Git, Node.js, LangChain, LangGraph, MLflow, FastAPI, postgreSQL, React
Cloud Platforms	GCP (Vertex AI), AWS, Azure Machine Learning
Other	LaTeX, Praat (Basic), HTML, CSS
Spoken Languages	English (Native), German (B1.1), Kannada (Native), Hindi (Fluent), French (Basic), Chinese (Basic)

Certifications

- **Machine Learning Engineering Professional Certificate (Google Cloud)** — Google Cloud, Coursera
- **IBM RAG and Agentic AI Specialization** — IBM, Coursera
- **Microsoft Azure for AI and Machine Learning** — Microsoft, Coursera
- **Deep Learning Specialization** — DeepLearning.AI, Coursera

Achievements & Other Projects

- **Winner - Best Visualization, AI + Environment EcoHackathon**
  ETH BioDivX, GainForest, ETH AI Center, 2024.
- **Winner - Coding Competitions, JSS Science & Technology University**
  Secured 1st and 2nd place in various competitive coding and problem-solving events.
- **Language consultant - ISLE**
  Worked as language consultant for linguistic elicitation class at UZH.

Interests

- **Creative Tech:** Blender, Unity 3D, Photoshop
- **Other:** Music, Sketching, Bouldering